##UK Food Price Intelligence: Detecting Volatility, Seasonality, and
Market Vulnerabilities in Fresh Produce

In [1]:
```
import pandas as pd
df = pd.read_csv("./fruitvegprices.csv")
```

In [2]:
```
df.head()
```

Out[2]:

| | category | item | variety | date | price | unit |
|---|---|---|---|---|---|---|
| 0 | fruit | apples | bramleys_seedling | 3/17/2025 | 1.35 | kg |
| 1 | fruit | apples | coxs_orange_group | 3/17/2025 | 1.26 | kg |
| 2 | fruit | apples | braeburn | 3/17/2025 | 1.17 | kg |
| 3 | fruit | apples | gala | 3/17/2025 | 1.24 | kg |
| 4 | fruit | apples | other_mid_season | 3/17/2025 | 1.37 | kg |

In [3]:
```
df.describe()
```

Out[3]:

| | price |
|---|---|
| count | 15917.000000 |
| mean | 1.716899 |
| std | 2.124775 |
| min | 0.000000 |
| 25% | 0.630000 |
| 50% | 1.000000 |
| 75% | 1.700000 |
| max | 20.000000 |

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15917 entries, 0 to 15916
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   category  15917 non-null  object
 1   item      15917 non-null  object
 2   variety   15917 non-null  object
 3   date      15917 non-null  object
 4   price     15917 non-null  float64
 5   unit      15917 non-null  object
dtypes: float64(1), object(5)
memory usage: 746.2+ KB
```

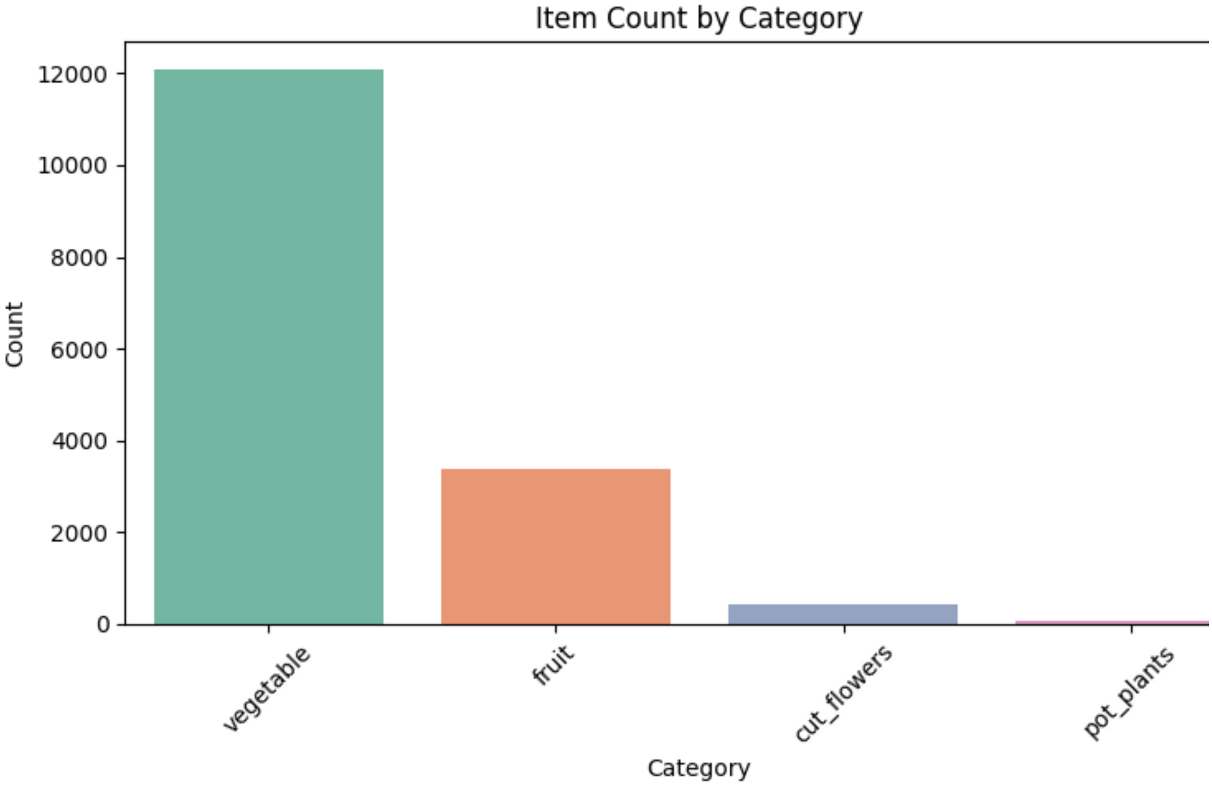
In [5]:
```
df.isnull().sum()
```

```
Out[5]: category    0
        item        0
        variety     0
        date        0
        price       0
        unit        0
        dtype: int64
```

```
In [6]: negative_or_zero_prices = df[df['price'] <= 0]
        negative_or_zero_prices
        df = df[df['price'] > 0]
```

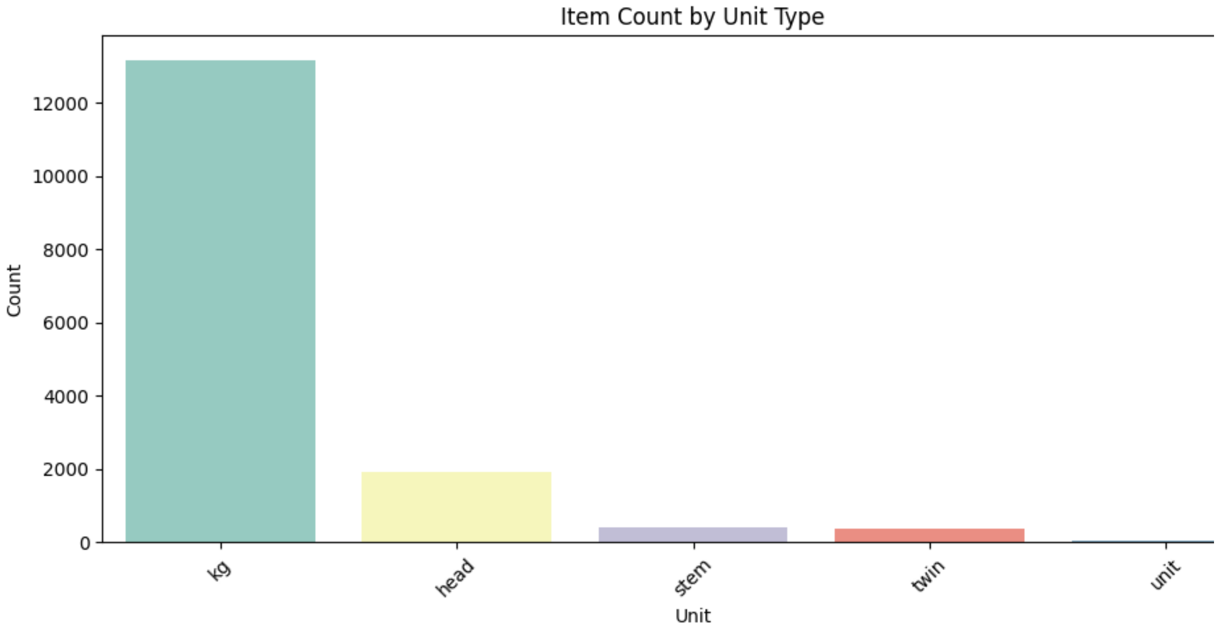
```
In [7]: import matplotlib.pyplot as plt
        import seaborn as sns

        total_categories = df['category'].nunique()
        print("total_categories",total_categories)
        category_counts = df['category'].value_counts()
        plt.figure(figsize=(8,5))
        sns.barplot(x=category_counts.index,  hue=category_counts.index,  y=catego
        plt.title("Item Count by Category")
        plt.xlabel("Category")
        plt.ylabel("Count")
        plt.xticks(rotation=45)
        plt.tight_layout()
        plt.show()
```

```
total_categories 4
```



In [8]:
```
total_unit = df['unit'].nunique()
print("total unit : ",total_unit)
unit_counts = df['unit'].value_counts()


plt.figure(figsize=(10,5))
sns.barplot(x=unit_counts.index,     hue=unit_counts.index,   y=unit_count:
plt.title("Item Count by Unit Type")
plt.xlabel("Unit")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
total unit :  5
```



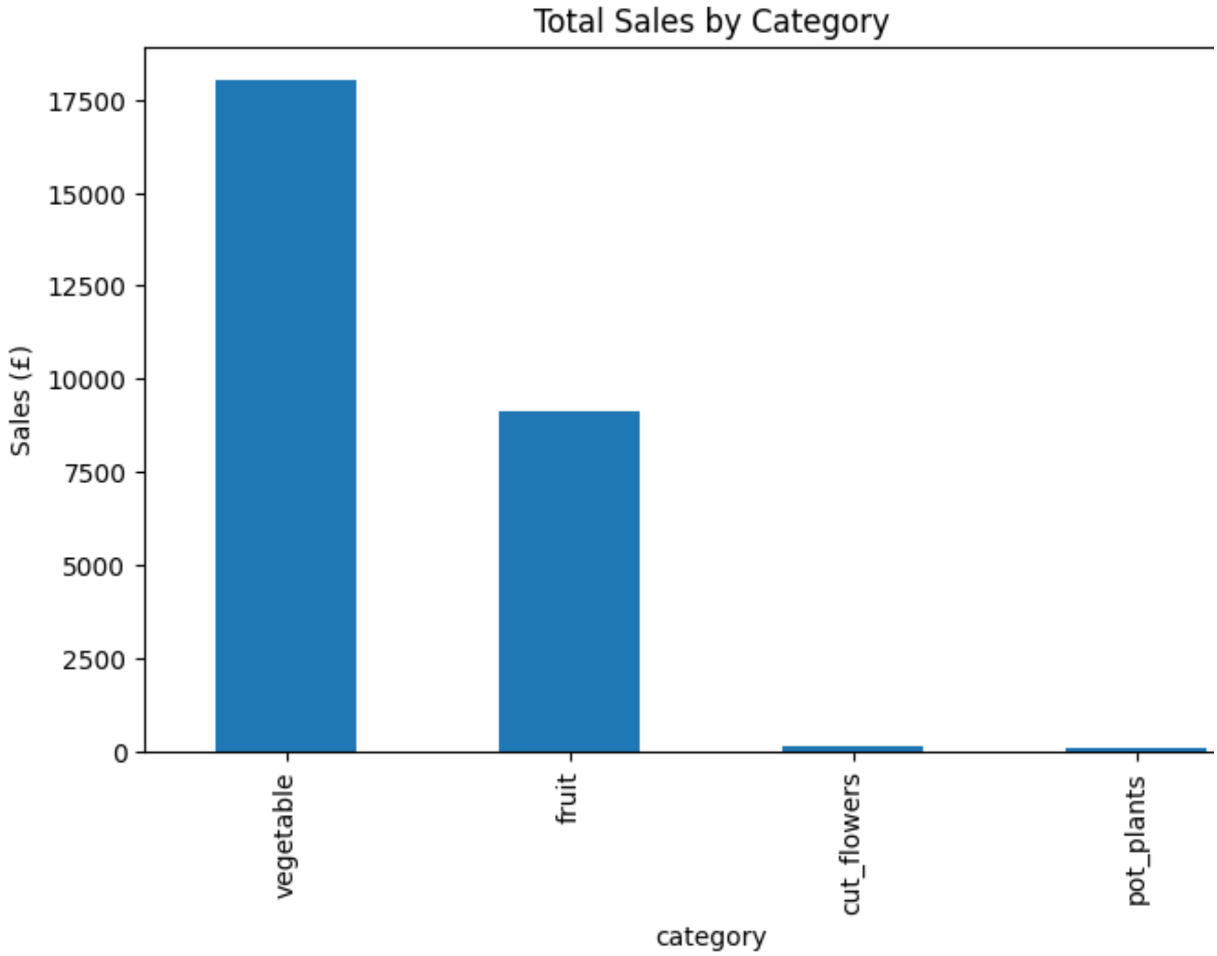

In [9]:
```
summary = df.groupby('category').agg(
    total_items=('price', 'count'),
    total_sales=('price', 'sum'),
    average_price=('price', 'mean'),
    max_price=('price', 'max'),
    min_price=('price', 'min'),
).sort_values(by='total_sales', ascending=False)

print(summary)
```

```
             total_items  total_sales  average_price  max_price  min_price
category
vegetable          12080     18020.58       1.491770      20.00       0.12
fruit               3367      9128.10       2.711048      18.85       0.22
cut_flowers          420       114.27       0.272071       0.99       0.02
pot_plants            49        64.93       1.325102       2.13       0.55
```

In [10]:
```
# Dropping pot_plants
# Remove rows where category is 'pot_plants'
df = df[df['category'].str.strip().str.lower() != 'pot_plants']
```

In [11]:
```
                                        ##Total sales by Each category
total_sales_by_category = df.groupby('category')['price'].sum().sort_valu
print("Total Sales by Category:\n", total_sales_by_category)
import matplotlib.pyplot as plt

summary['total_sales'].plot(kind='bar', title='Total Sales by Category', y
plt.show()
```

```
Total Sales by Category:
 category
vegetable      18020.58
fruit           9128.10
cut_flowers      114.27
Name: price, dtype: float64
```




In [12]:
```
                                        ##Average Price by Category
avg_price_by_category = df.groupby('category')['price'].mean().sort_values
print("Average Price by Category:\n", avg_price_by_category)
```

```
Average Price by Category:
 category
fruit          2.711048
vegetable      1.491770
cut_flowers    0.272071
Name: price, dtype: float64
```

In [13]:
```
df.head()
```

Out[13]:

| | category | item | variety | date | price | unit |
|---|---|---|---|---|---|---|
| **0** | fruit | apples | bramleys_seedling | 3/17/2025 | 1.35 | kg |
| **1** | fruit | apples | coxs_orange_group | 3/17/2025 | 1.26 | kg |
| **2** | fruit | apples | braeburn | 3/17/2025 | 1.17 | kg |
| **3** | fruit | apples | gala | 3/17/2025 | 1.24 | kg |
| **4** | fruit | apples | other_mid_season | 3/17/2025 | 1.37 | kg |

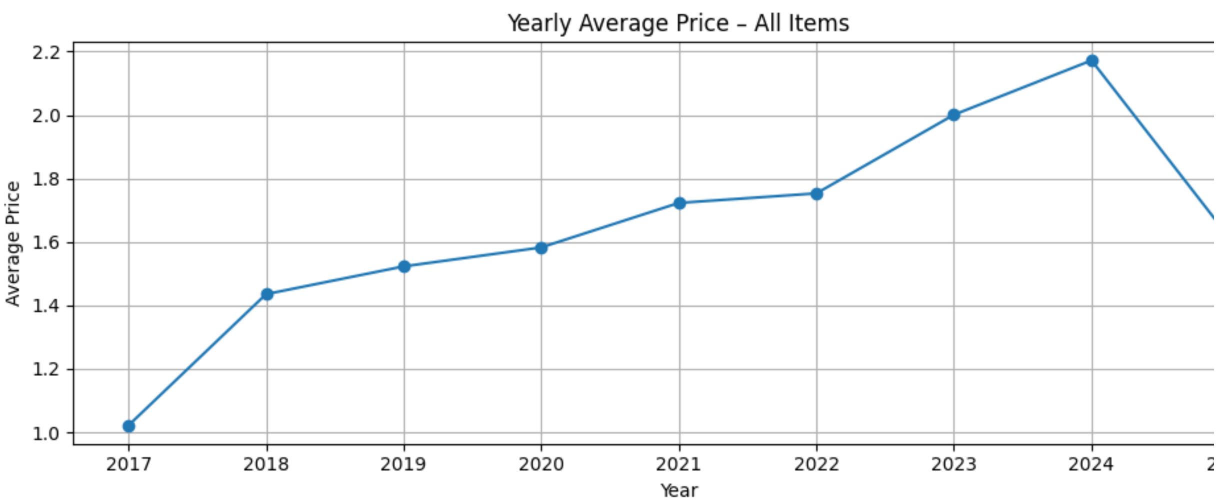
In [14]:

```
df.head()
```

Out[14]:

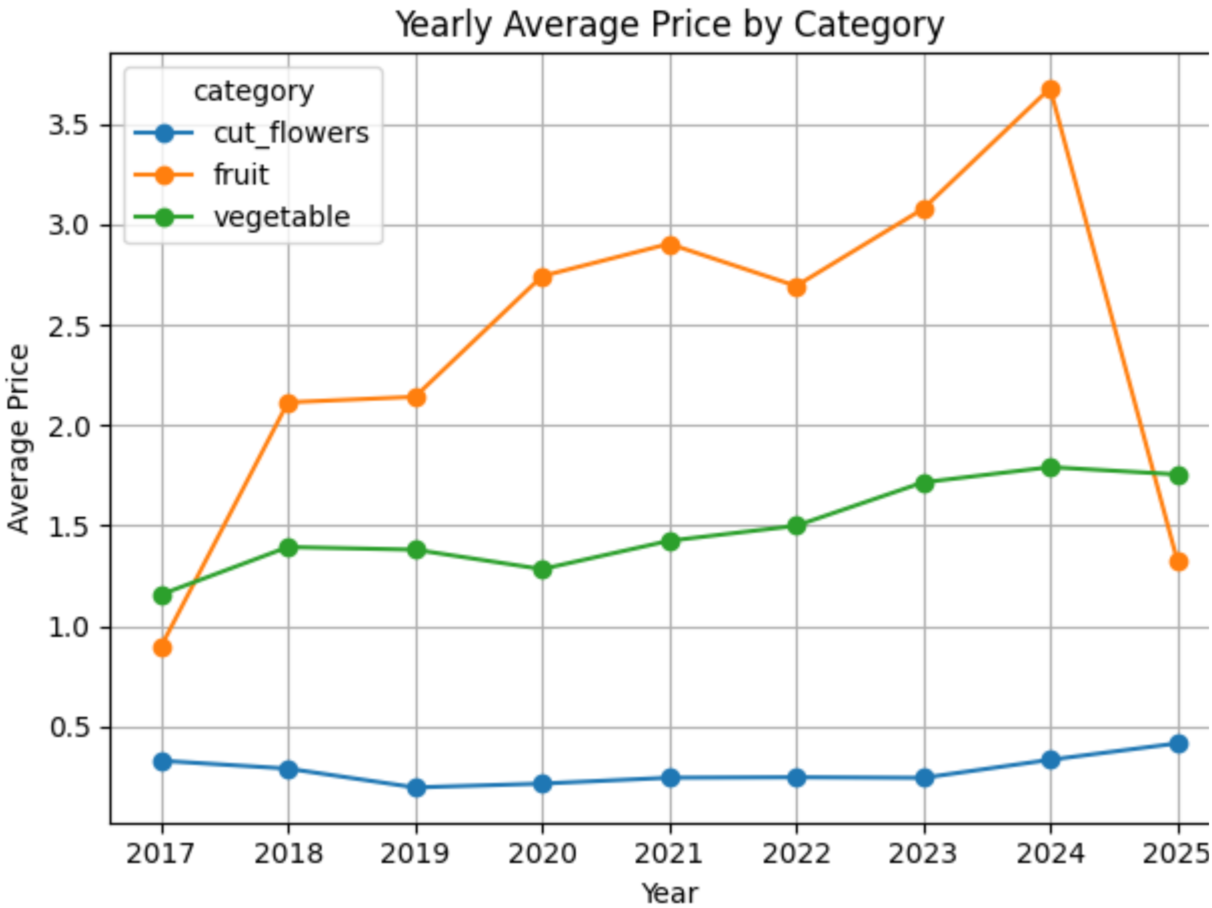
| | category | item | variety | date | price | unit |
|---|---|---|---|---|---|---|
| **0** | fruit | apples | bramleys_seedling | 3/17/2025 | 1.35 | kg |
| **1** | fruit | apples | coxs_orange_group | 3/17/2025 | 1.26 | kg |
| **2** | fruit | apples | braeburn | 3/17/2025 | 1.17 | kg |
| **3** | fruit | apples | gala | 3/17/2025 | 1.24 | kg |
| **4** | fruit | apples | other_mid_season | 3/17/2025 | 1.37 | kg |

```
In [15]: import pandas as pd
         import matplotlib.pyplot as plt
         import seaborn as sns
         import warnings

         warnings.filterwarnings('ignore')

         # === 0. Make a copy & clean column names ===
         df = df.copy()
         df.columns = df.columns.str.strip()

         # === 1. Ensure 'date' is a column, not just an index ===
         if not 'date' in df.columns:
             # if your df.index is a DatetimeIndex, bring it back as a column
             if isinstance(df.index, pd.DatetimeIndex):
                 df = df.reset_index().rename(columns={'index':'date'})
             else:
                 raise KeyError("No 'date' column found and index is not datetime.'

         # === 2. Convert and extract date parts ===
         df['date'] = pd.to_datetime(df['date'])
         df['year']      = df['date'].dt.year
         df['month_num'] = df['date'].dt.month
         df['month']     = df['date'].dt.strftime('%b')  # Jan, Feb, …

         # === 3. (Optional) Drop unwanted category ===
         df = df[df['category'].str.lower().str.strip() != 'pot_plants']

         # === 4. Yearly average price – all items ===
         yearly_avg_all = df.groupby('year')['price'].mean()

         plt.figure(figsize=(10,4))
         yearly_avg_all.plot(marker='o')
         plt.title('Yearly Average Price – All Items')
         plt.xlabel('Year')
         plt.ylabel('Average Price')
         plt.grid(True)
         plt.tight_layout()
         plt.show()
```

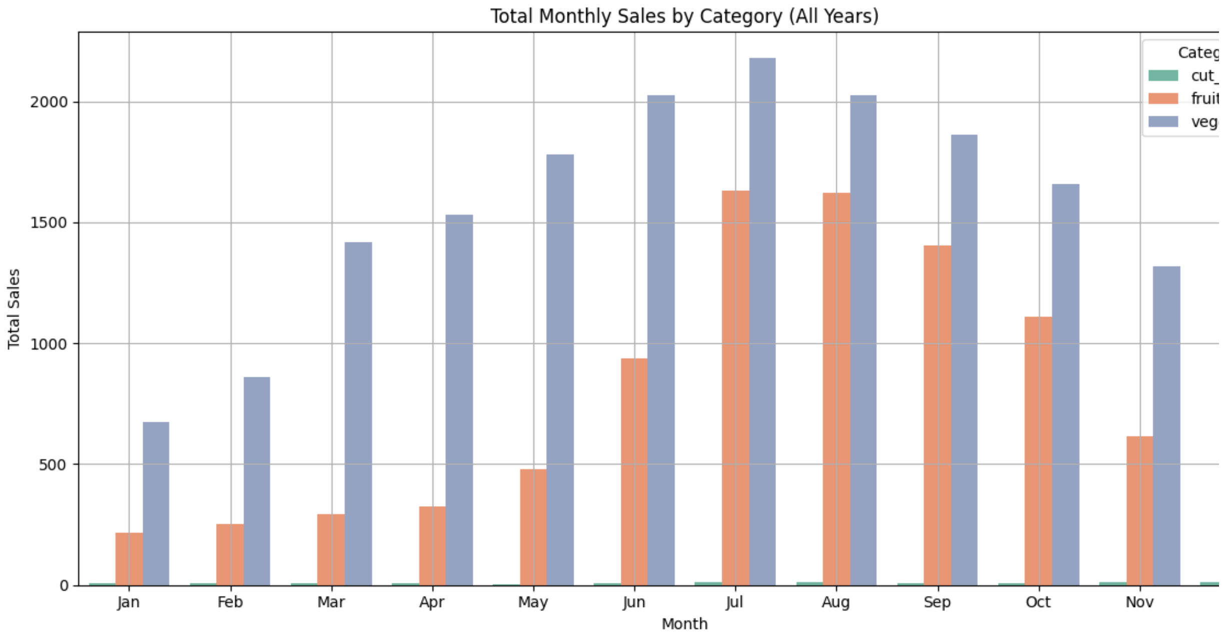


In [16]:
```
# === 5. Yearly average price — by category ===
yearly_avg_cat = df.groupby(['year','category'])['price'].mean().unstack(

plt.figure(figsize=(10,5))
yearly_avg_cat.plot(marker='o')
plt.title('Yearly Average Price by Category')
plt.xlabel('Year')
plt.ylabel('Average Price')
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```



In [17]:
```python
# === 6. Total monthly sales by category (all years) ===
monthly_sales = (
    df
    .groupby(['month_num','month','category'])['price']
    .sum()
    .reset_index()
    .sort_values('month_num')
)

plt.figure(figsize=(12,6))
sns.barplot(
    data=monthly_sales,
    x='month', y='price',
    hue='category', palette='Set2',
    order=['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','N
)
plt.title('Total Monthly Sales by Category (All Years)')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.xticks(rotation=0)
plt.legend(title='Category')
plt.grid(True)
plt.tight_layout()
plt.show()
```




In [18]:
```python
import pandas as pd

# Step 2: Group by category and item, then count the occurrences
item_counts = df.groupby(['category', 'item']).size().reset_index(name='c

# Step 3: For each category, get the most common item (highest count)
most_common_items = item_counts.loc[item_counts.groupby('category')['coun

# Step 4: Display the result
print(most_common_items)
```

```
       category     item  count
8   cut_flowers   tulips    141
9         fruit   apples   1789
23    vegetable  cabbage   1669
```

```
In [19]: df[df['category'].str.lower().str.strip() == 'fruit']['item'].unique()
```

```
Out[19]: array(['apples', 'pears', 'strawberries', 'blackberries', 'raspberries',
                'blueberries', 'plums', 'cherries', 'currants', 'gooseberries'],
               dtype=object)
```